Brainwave Architecture
Hardware Aspects and Evaluation Results

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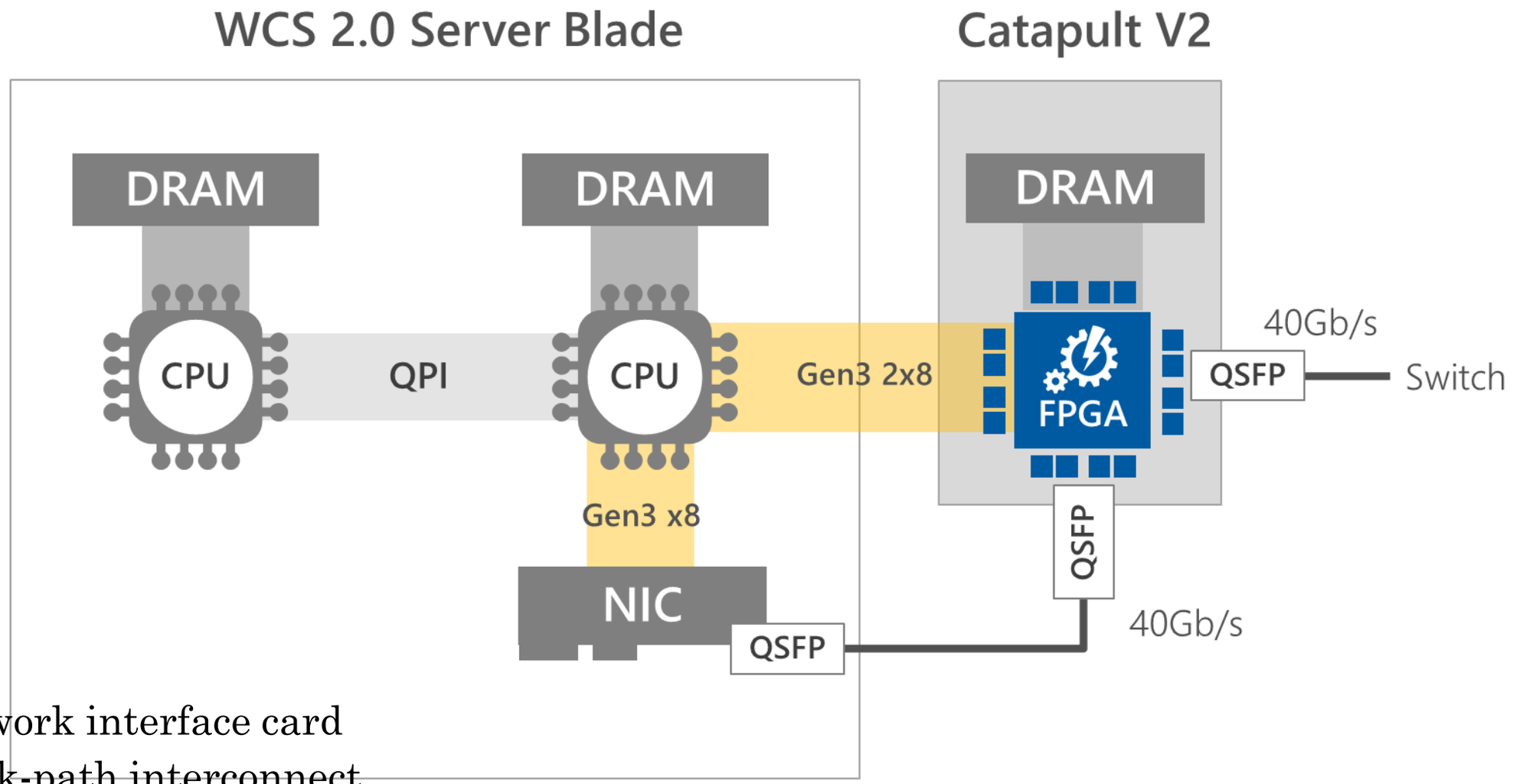
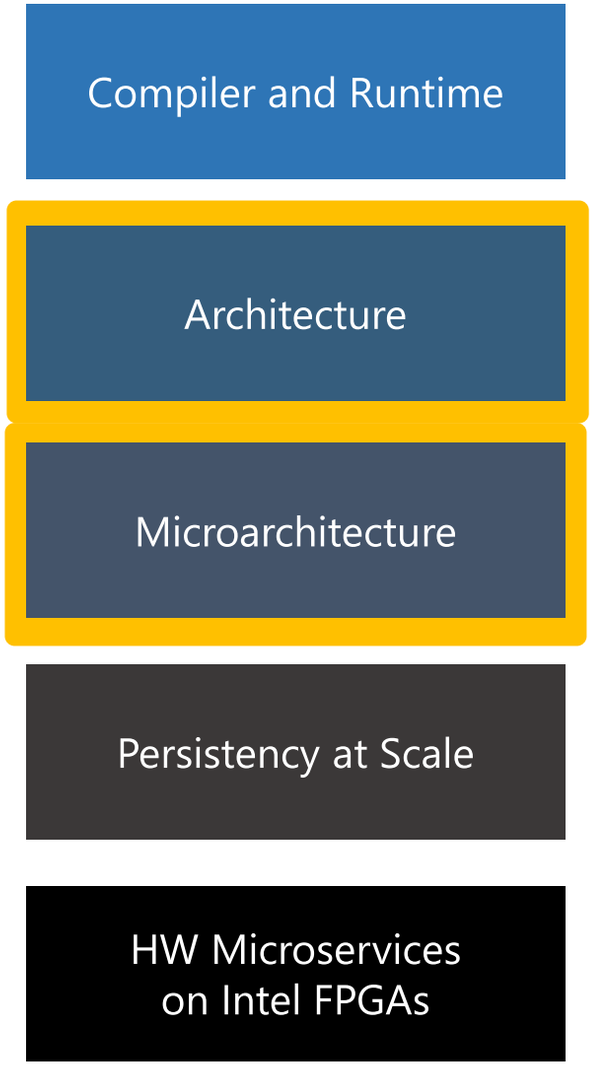
All images are courtesy of: http://learningsys.org/nips17/assets/slides/brainwave-nips17.pdf and https://www.servethehome.com/microsoft-shows-off-project-brainwave-persistent-inferencing-from-fpga-cache/

FPGAs Deployed in Microsoft Servers Worldwide



NIC—network interface card
QPI—quick-path interconnect
QSFP—quad small form-factor pluggable

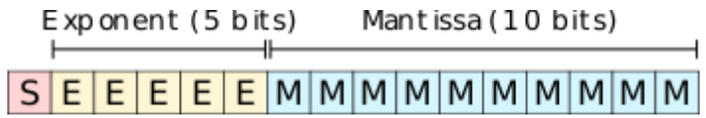
The BrainWave Stack



Brainwave NPU features

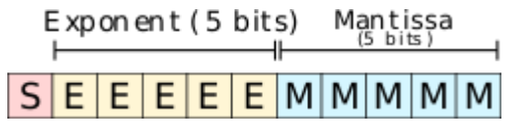
- The use of compile-time narrow precision data types to extract higher performance than what is possible using conventional float and integer types without loss in accuracy.

IEEE 754 half-precision floating point FP16

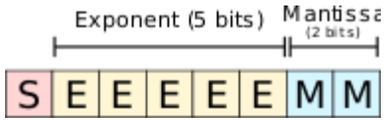


Microsoft's proprietary parameterizable narrow precision format wrapped in FP16 interfaces

MSFP11

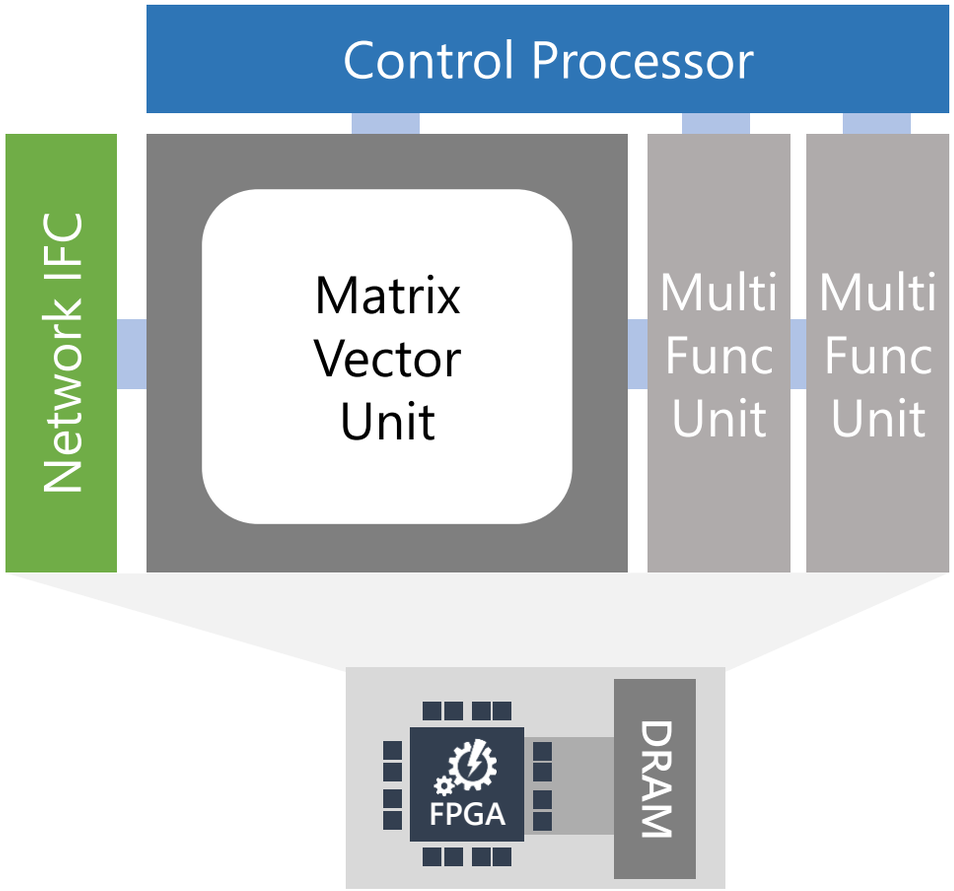


MSFP8



BrainWave Soft DPU Architecture

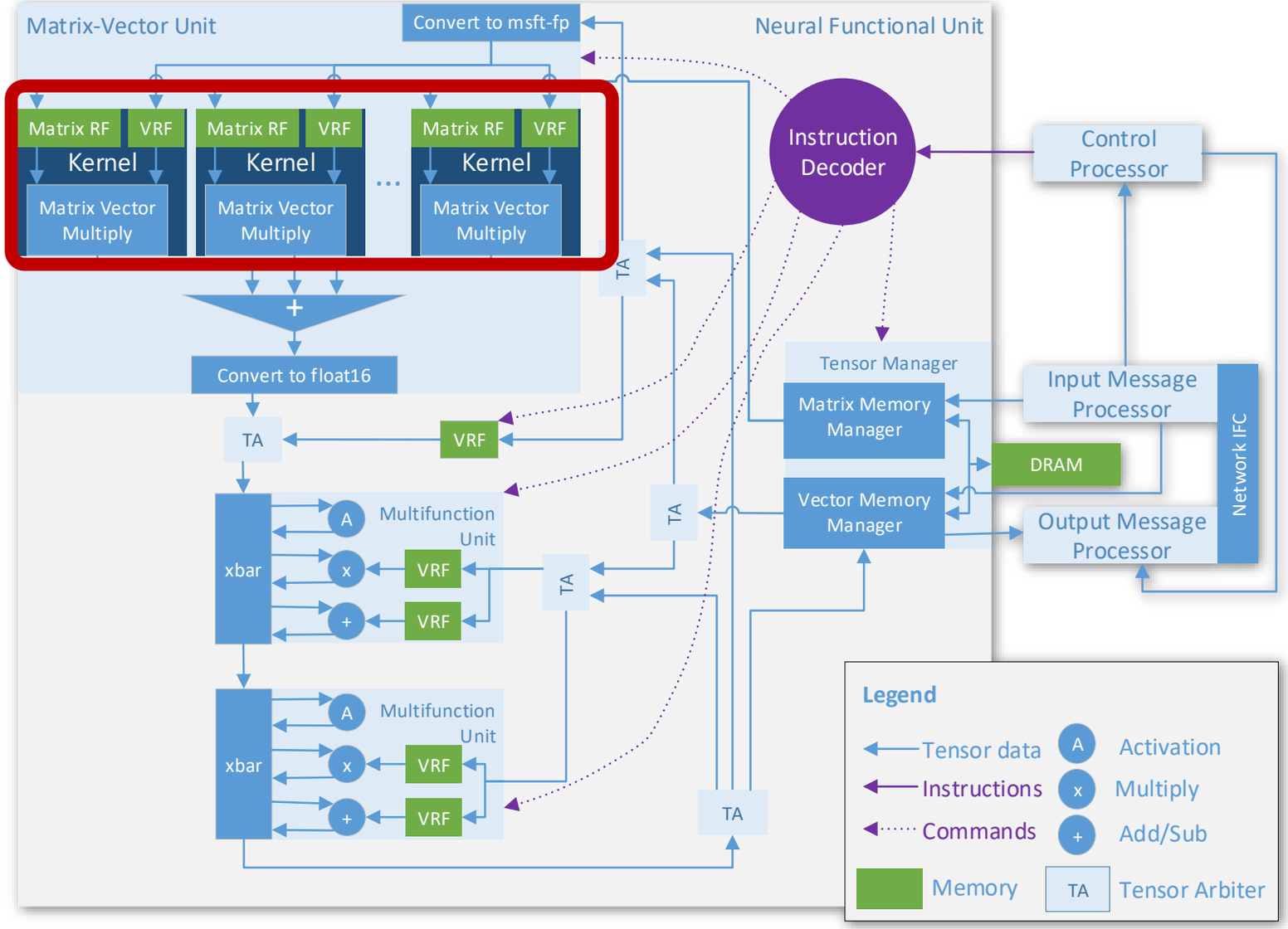
- Single-threaded C programming model (no RTL).
- ISA with specialized instructions: dense matrix multiplication, convolutions, nonlinear activations, vector operations, and embeddings.
- Multifunction units perform different vectorized functions, such as sigmoid and dot products.
- Microarchitecture is parameterizable and scales to large FPGAs, e.g., those with ~1 million adaptive logic modules.



ISA = Instruction Set Architecture,
RTL = Register Transfer Level

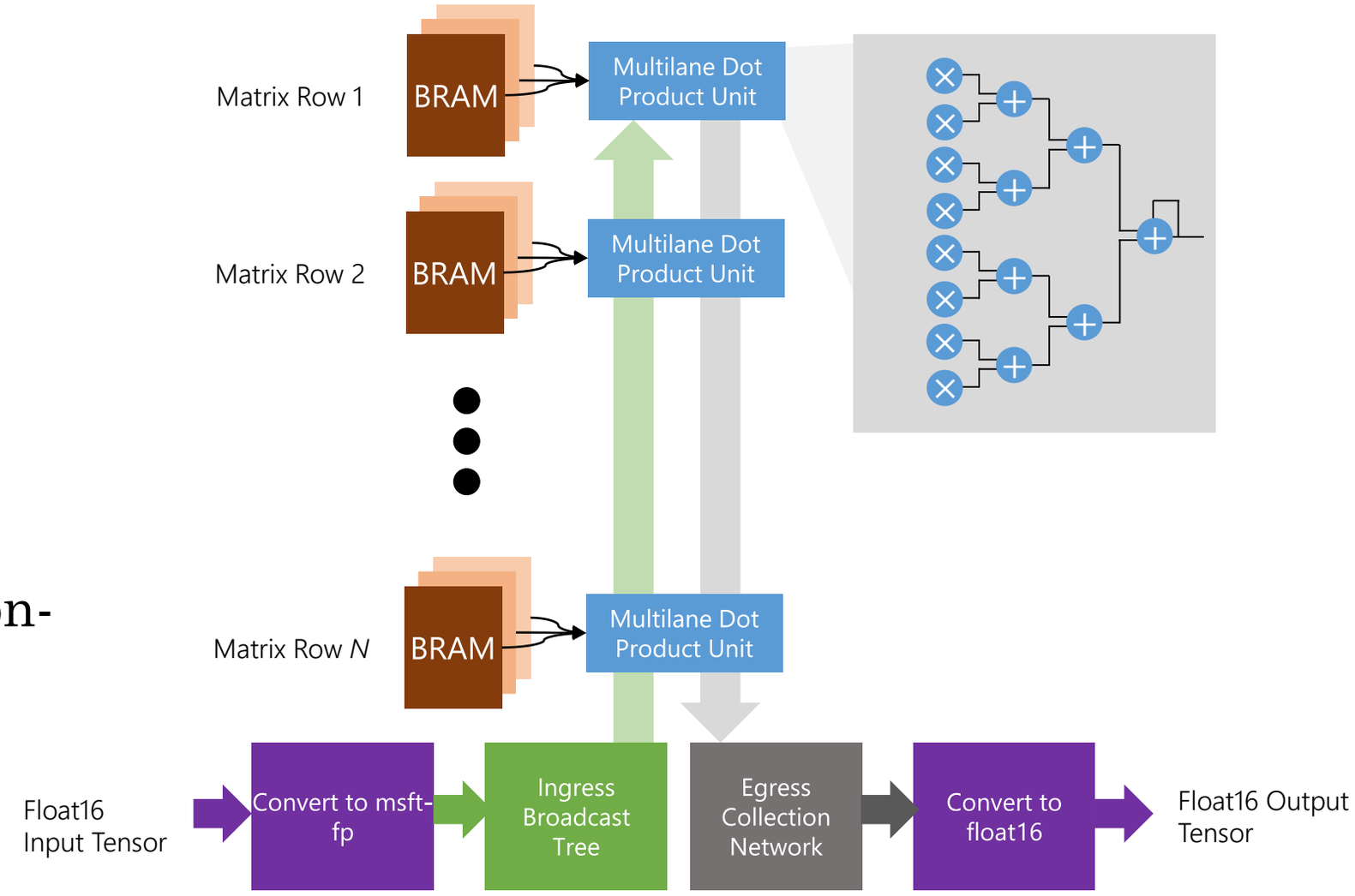
BrainWave Soft DPU Microarchitecture

- Optimized for matrix vector multiplication with batch size of 1
- Dense dot product units map efficiently to soft logic and DSPs

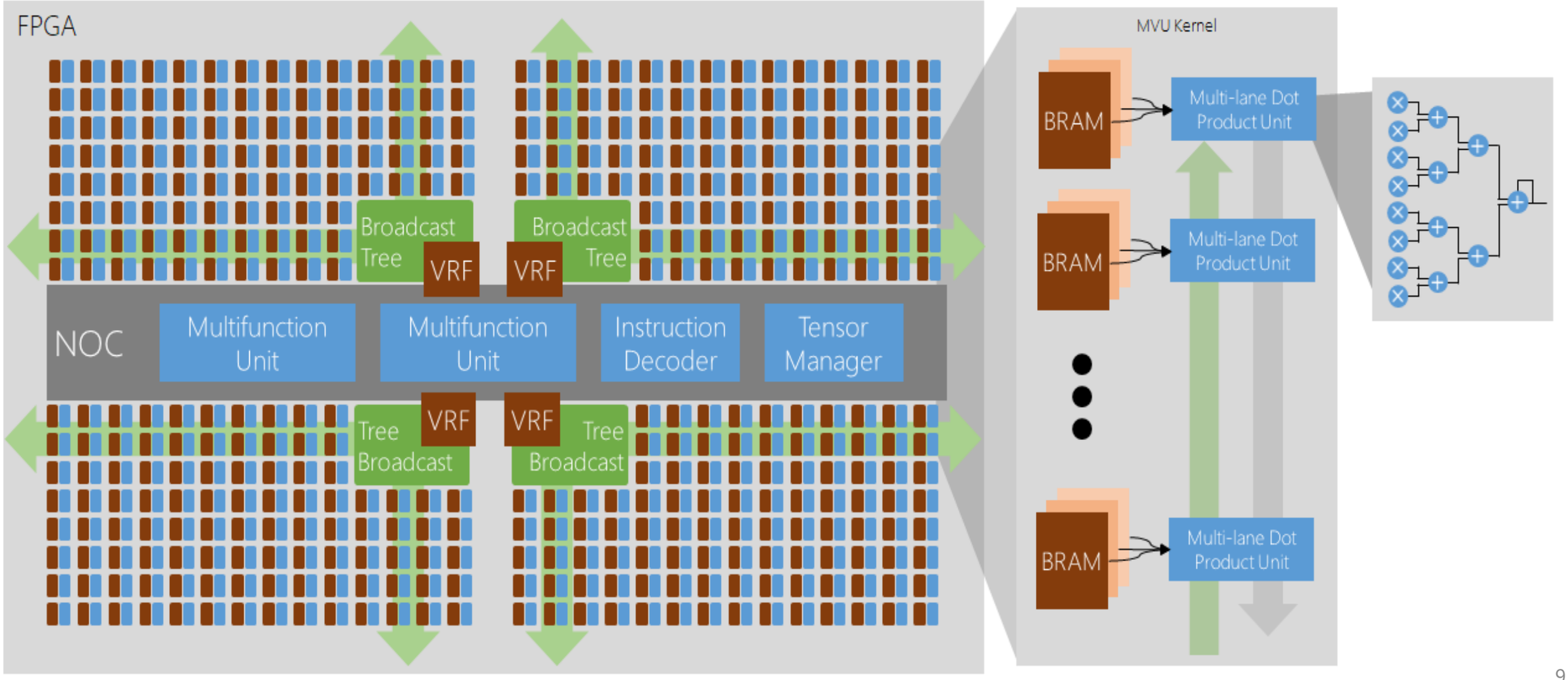


Matrix Vector Multiply Unit

- Matrices distributed row-wise across 1–10K banks of BRAM, bandwidth up to 20 TB/s
- Locally attached SRAMs feed each MAC with unique weight values, enabling full-chip utilization of MVM operations
- Can scale to use all available on-chip BRAMs, DSPs, and soft logic
- *In situ* conversion of float16 weights and activations to internal format



Matrix Vector Multiply Unit



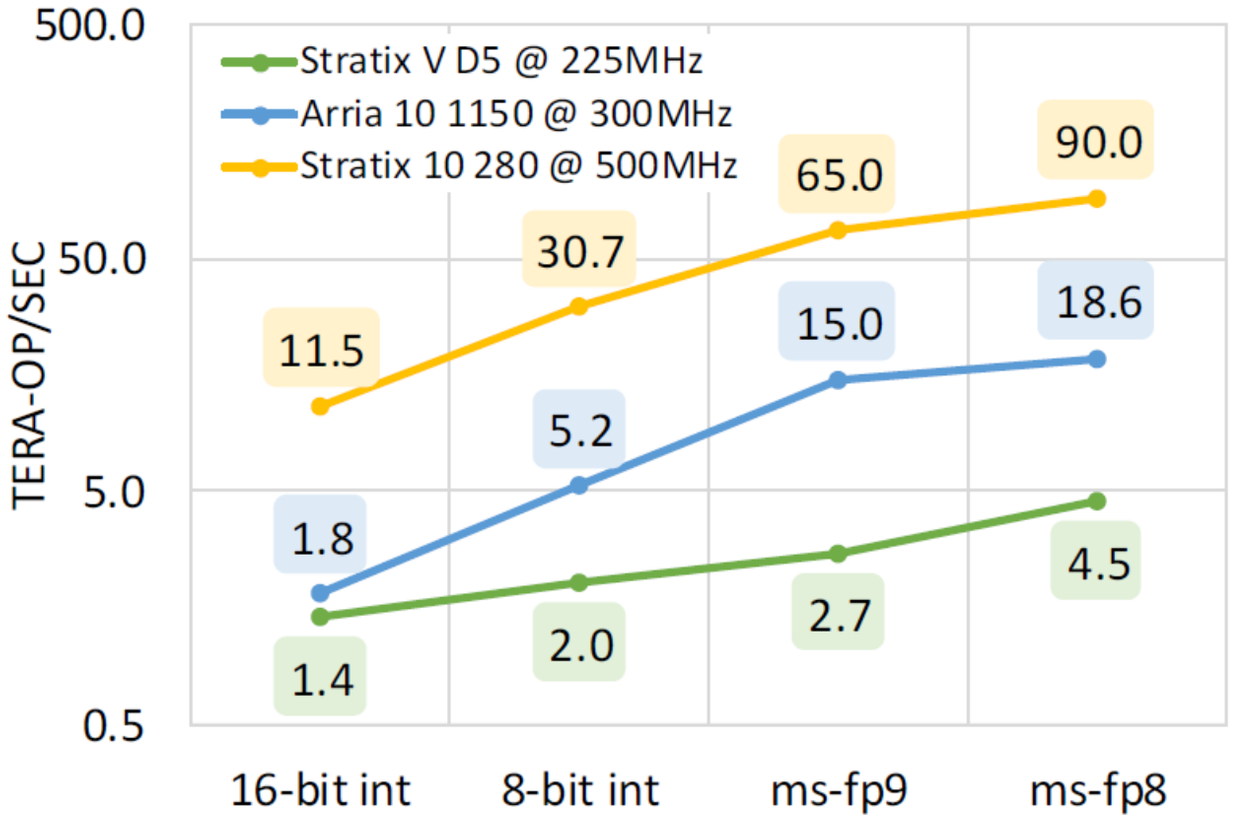
Evaluation Results

Evaluation Hardware

- Stratix V FPGA: 28 nm
- Arria 10 1150 FPGA
- Stratix 10 280 FPGA: 14 nm

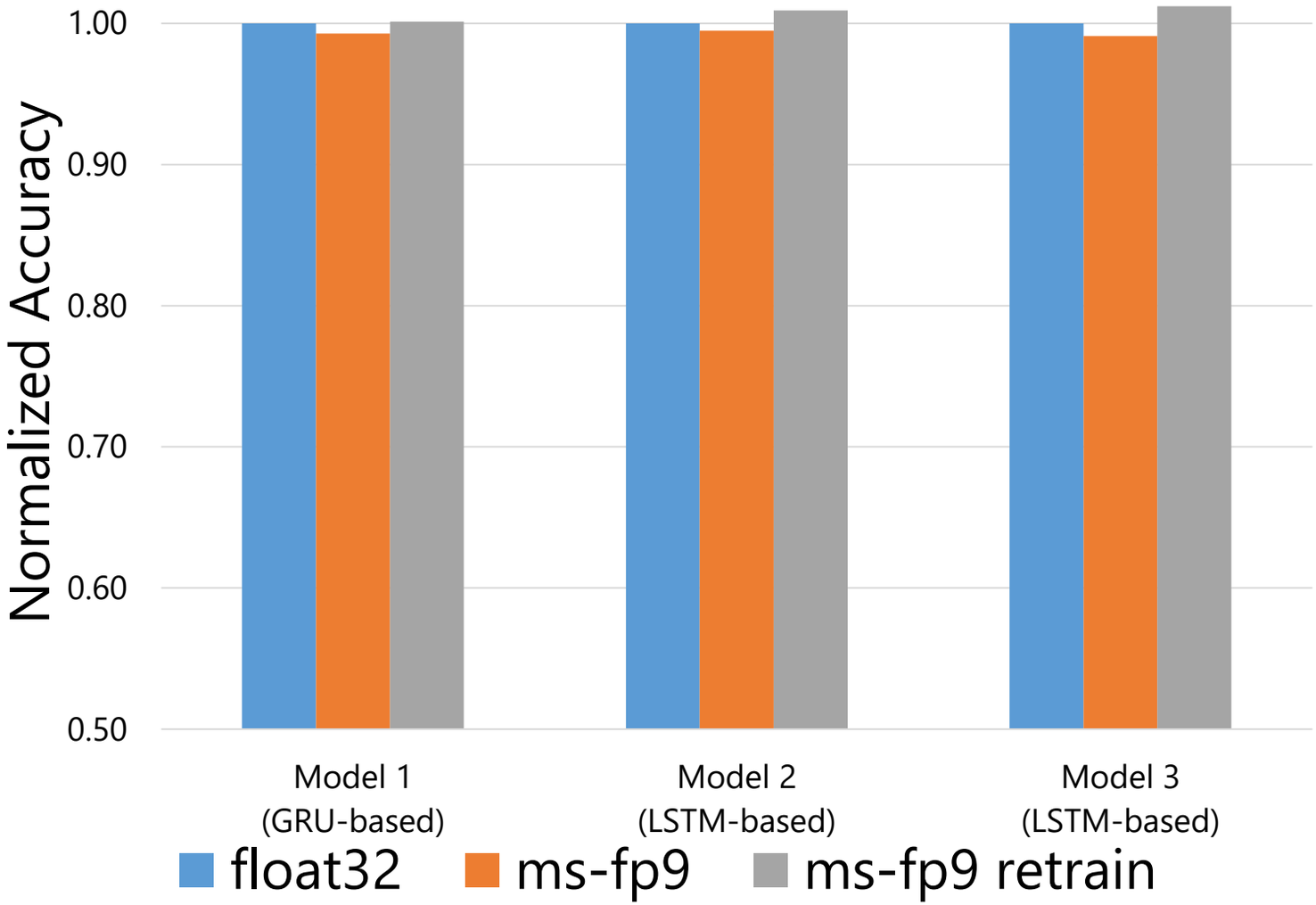
- Compared to Stratix V D5 FPGA, Stratix 10 280 has 3× higher clock speed, 5.4× more soft logic resources, 3.6× more DSPs, and 5.8× additional on-chip memory.

Peak Performance of Brainwave Across Three Generations of FPGAs



- Use of ms-fp8 narrow precision improves performance by 3.2× to 7.8× over a conventional 16-b fixed point.
- Here, batch size is 1 ⇒ result is provided in real time.
- If batch size were to be large, we would have to wait for each batch to finish before we can get an answer.

Impact of Narrow Precision on Accuracy



Results on Bing

TP—Turing prototype

Bing TP1			
	CPU-only	**Brainwave-accelerated**	**Improvement**
Model details	GRU 128x200 (x2) + W2Vec	LSTM 500x200 (x8) + W2Vec	Brainwave-accelerated model is > 10X larger and > 10X lower latency
End-to-end latency per Batch 1 request at 95%	9 ms	0.850 ms	
Bing DeepScan			
	CPU-only	**Brainwave-accelerated**	**Improvement**
Model details	1D CNN + W2Vec *(RNNs removed)*	1D CNN + W2Vec + GRU 500x500 (x4)	Brainwave-accelerated model is > 10X larger and 3X lower latency
End-to-end latency per Batch 1 request at 95%	15 ms	5 ms	

Chung, Eric, et al. "Serving DNNs in real time at datacenter scale with project brainwave." IEEE Micro 38.2 (2018): 8-20.

Comments

- To meet end-to-end latency requirements, CPU-only versions are scaled down in both parameters and operations by an order of magnitude relative to Brainwave-accelerated versions.
- Scaled down TP1 incurs 9 ms on Bing's CPU-only infrastructure.
- Brainwave enables serving of a larger and more accurate bidirectional LSTM variant of the model, with hidden states increased from 128 to 500.
- Brainwave model size is > 10× of CPU-only model and > 10× lower latency at 0.85 ms.
- Space–time speedup of Brainwave >100×.

Conclusions

- Project BrainWave is a powerful platform for an accelerated AI cloud.
  - It runs on Microsoft's hyperscale infrastructure with FPGAs.
  - It achieves excellent performance at low batch sizes via persistency and narrow precision.
  - It is adaptable to precision and changes in future AI algorithms.

Reference

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